Fiscal Year:	FY 2010	**Task Last Updated:**	FY 11/03/2010
PI Name:	McGinnis, Michael R Ph.D.		
Project Title:	PharmaSat: Effect of Microgravity upon Yeast Susceptibility to Antifungal Drugs		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	None		
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Zip Code:	77555 - 0609	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Directed Research
Start Date:	02/01/2007	**End Date:**	01/31/2010
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Agasid, Elwood	**Contact Phone:**	650-604-0558
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Flight Program:	Shuttle		
Flight Assignment:	STS-123 Small Satellites NOTE: PharmaSat flight delayed ; end date chgd to 1/31/2010 (4/2009) NOTE: NCE to 1/31/2009 per A. Chu/ARC; original end date was 1/31/2008 (6/08)		
Key Personnel Changes/Previous PI:	NOTE: David Niesel added as CoInvestigator in late 2009, per ARC (ed. 3/10)		
COI Name (Institution):	Niesel (as of 9/2009), David (University of Texas)		
Grant/Contract No.:	NNX07AD28A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The PharmaSat experiment was designed to measure the effect of microgravity upon yeast resistance to the antimicrobial agent - voriconazole. Voriconazole is a stable and broad spectrum antimicrobial that targets the fungal cytochrome p450 sterol 14-demethylase enzyme (CYP51) Previous experimentation in microgravity and under modeled microgravity conditions suggest that in these environments, resistance to azoles is increased. Voriconazole has excellent stability under ambient conditions and exhibits a broad spectrum of activity and is being evaluated in vitro as a potential antifungal counter-measure in space. The PharmaSat experimental design uses a recognized laboratory testing method for antimicrobial resistance. The experiment investigated three concentrations of voriconazole (a low concentration, a concentration at the minimal inhibitory concentration, MIC and a high dose, plus a no-drug control) for their effect on yeast growth cell and viability. Saccharomyces cerevisiae was used as a model microorganism and has been the subject of previous space-flight studies, and is used as a standard to evaluate resistance to antifungal agents. Growth was measured by two independent methods - optical density (OD) and Alamar Blue reduction as a metabolic indicator. Statistical analysis of the data used longitudinal data, a more robust approach than traditional end-point-based analysis. The underlying hypothesis for this experiment was “yeast growing in microgravity has altered resistance to azole antifungal agents”. The experiment was conducted in a 59-well Fluidic Card with 48 biology wells in four independently plumbed banks with 11 reference wells in the center bank. Gas- permeable membranes on both sides of the wells allowed for air exchange and 1µm Nylon filter membranes retained the yeast cells inside the wells The experiment used a complex fluidic system capable of two fluid exchanges, dilution of the antifungal agent and replenishing evaporative losses by using 2 pumps, 9 reservoir bags and 14 valves.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Overall, this study revealed new information about yeast resistance to antimicrobial agents. Specifically, the PharmaSat experiment was designed to measure the affect of microgravity upon yeast resistance to the antifungal agent voriconazole. Previous studies in microgravity or modeled microgravity suggest that resistance to azoles is increased. The data from the PharmaSat mission advances our understanding of microbial pathogenic mechanisms when these organisms adapt to different environments. This is important to understand in the space environment and on Earth. Understanding how fungi and other microbes respond to the space environment and become more resistant to therapeutics will additionally allow a rationale risk assessment for this and other microbial pathogens. This will contribute to the success of long-term space missions and for the habitation of space. These results advance NASA's goals of protecting human health during long duration space travel and potentially provide information that could lead to development of countermeasures - to detect illnesses, protect individuals from infectious diseases or treat these diseases in space. Additionally, the new information obtained in this study will have direct earth benefits in terms of understanding the diseases and drug resistance mechanisms and could lead to novel vaccines and therapeutics for microbial pathogens.
Task Progress:	PharmaSat was launched on a Minotaur I rocket, in May, 2009. The nanosatelite was deployed ~20 min later and radio contact was established during the first orbit. The satellite orbit became stabilized in ~24 hours. Its orbital trajectory put PharmaSat in sunlight ~ 67% and darkness ~ 33% of each 97-min orbit. After stabilization (gravity < 10-3 g), at t = 47 hr after orbital deployment, the temperature of the fluidic card was set to 27 °C. After allowing 3.6 hr for temperature stabilization, S. cerevisiae growth was initiated (in all 48 wells of the microfluidic plate) by introducing RPMI1640 cell culture medium containing the viability dye - Alamar blue. Each 12-well bank of wells was activated in sequence. Two concentrations (0.1 and 0.2 OD600) of the yeast were utilized. In each bank one set of six wells were loaded with yeast at the lower optical density, while the remaining 6 wells of each bank were loaded with yeast at the higher OD (0.2). This strategy insured that if insufficient organisms survived the stasis period in the lower OD groups of wells, the higher OD wells would provide an ~2X higher initial population to begin growth. Since antifungal tests are more reliable at lower cell densities, the successful growth of all 24 wells at the lower OD cultures led us to choose those data for our initial ground/flight comparisons. Voriconazole, a stable antifungal agent was introduced after the yeast had initiated growth. Based on initial growth rates, voriconazole was introduced by pumping ~ 12 hr after initial growth medium introduction. Ground controls for this experiment consisted of an identical satellite system at NASA Ames, housed in a thermal chamber with temperatures set to those reported by the satellite. This Ground Control, with S. cerevisiae loaded from the same flight cultures was conducted as a “delayed synchronous” (~1 day) ground control experiment to compare ground and space results. After allowing for initial growth (re-activation of the yeast cultures), growth medium + voriconazole was pumped into the four 12-well banks at 0.25X, 1X, and 4X the minimum inhibitory concentration (MIC). The MIC of voriconazole for S. cerevisiae was 0.5 µg/mL in ground studies. This allowed each of the four conditions tested with multiple independent microwells at both concentrations. Yeast cultures showed a longer “lag time” (time to the mid-point of mid-logarithmic growth), growth time (time in exponential growth) and doubling time in space samples than ground controls. All values were significant except for cell division for the growth time measurement. It may be the case that the yeast grow more slowly in space because the transport of nutrients to the cells and/or the transport of waste products away from the cells is slower, being driven solely by diffusion, than on Earth where convection can play a role in mass transport. Overall, the PharmaSat experiment was a success. Cell growth and metabolic activity was observed in all wells, and an antifungal effect was observed in both ground and spaceflight samples. All equipment functioned nominally. Biological results showed: 1) differences in the growth rates between flight and ground samples, 2) antifungal activity was observed in both cultures, 3) no significant differences between the spaceflight and ground cultures were observed with the low and MIC concentrations but a large difference was observed at the high antifungal concentration.
Bibliography Type:	Description: (Last Updated: 11/13/2013)
Abstracts for Journals and Proceedings	Parra M, Ly D, Ricco AJ, McGinnis MR, Niesel D. "The PharmaSat Nanosatellite Platform for Life Science Experimentation: Effects of Space Flight on Antifungal Activity Against Saccharomyces cerevisiae." Paper presented at ASGSB Annual Meeting, Raleigh, NC, November 2009. Gravitational and Space Biology 2009 Nov;23(1):30. , Nov-2009

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Papers from Meeting Proceedings	Ricco AJ, Parra M, Niesel D, Ly D, Kudlicki A, McGinnis M, Hines JW. "Results from the PharmaSat Nanosatellite Mission: Dose Dependence of Growth and Metabolic Parameters for S. cerevisiae Grown in Microgravity and Challenged by Voriconazole." COSPAR Scientific Assembly, Bremen, Germany, July, 2010. COSPAR Scientific Assembly, Bremen, Germany, July, 2010. , Jul-2010